Dhruv Arora

Madison, WI, United States, 53703 | (602) 330-6880

dhruvarora0212@gmail.com | LinkedIn | GitHub | Website

CAREER OBJECTIVE

Recent graduate with a double major in Computer Science and Data Science with a Certificate in Economic Analytics from University of Wisconsin–Madison (May 2026). Experienced in data science, AI/ML, and software development, with hands-on work in LLM training and full-stack systems. Seeking a full-time role in software engineering, data, or applied AI.

SKILLS

- **Programming Languages**: Java, Python, C++, MATLAB, R
- **Web Development:** HTML, CSS, JavaScript, TypeScript, React
- **Databases:** SQL, SQLite, MySQL, JDBC
- **DevOps and Tools**: VS Code, Eclipse, IntelliJ, Git, Docker
- **Machine Learning & AI**: Spark, TensorFlow, Scala, PyTorch
- **Analytical Skills**: Problem Solving, Mathematical Reasoning, Data Analysis, Logical Reasoning
- **Soft Skills**: Focus, Communication, Collaboration, Adaptability

EDUCATION

B.S. in Computer Science and Data Science (Double Major), Certificate in Economic Analytics

University of Wisconsin-Madison, 2022 - 2026 | Cumulative GPA : 3.5

Coursework – Data Structures & Algorithms, Artificial Intelligence, Software Engineering, Database Management Systems, Data Science Modelling, Econometrics, Computer Engineering, Machine Organization, Human-Computer Interaction, Linear Algebra, Discrete Maths

EXPERIENCE

QUALCOMM (Industry-Sponsored Capstone Project), Madison, WI (*ML Engineering*) *January 2026 – Present*

- Built an Android app (Kotlin) recommending songs via a 6-stage on-device multimodal ML pipeline fusing text (MobileBERT), vision (EfficientNet-B0, int8 quantized), and sensor embeddings, ranked against pre-computed Spotify embeddings by cosine similarity.
- Deployed TensorFlow Lite models with NNAPI delegation for Qualcomm Hexagon NPU acceleration on Snapdragon devices, eliminating server dependency; implemented fallback encoders to maintain app functionality when TFLite assets are unavailable.
- Authored a 12-class evaluation suite covering latency, accuracy, memory, and pipeline correctness; integrated OAuth-based Spotify access for offline song embedding generation.

CENCORA (formerly AmerisourceBergen), Conshohocken, PA *(Data Engineering Intern)* *June 2025 – August 2025*

- Automated SAP spool parsing in Python for a 146+ TB SAP implementation, standardizing inconsistent log formats (plain-text and box-style) and cutting manual effort by ~60%.
- Integrated Delta Lake ETL pipelines in Databricks to optimize processing speed and reduce storage overhead; orchestrated scheduled runs with Airflow and added SQL-based validation checks for timely, accurate ingestion.
- Built Power BI dashboards visualizing row counts and date-of-write trends across the largest archiving objects, enabling stakeholder-driven prioritization of archival workflows.

UW–MADISON CDIS, Madison, WI *(Undergraduate Research Assistant – LLM Training)* *Aug 2025 – Present*

- Verified and analyzed AI-generated web navigation tasks, tracking step sequences, parallelization patterns, and URL trajectories to assess retrieval complexity.
- Refined ill-posed prompts and produced structured JSON datasets to ensure reproducible data for LLM training and evaluation.

WIPRO LIMITED, India (*Full Stack Developer Intern)* *June 2024 – August 2024*

- Developed and optimized Java 8 applications (Employee Management System & eShop) with JDBC/MySQL, improving performance, data reliability, and CRUD efficiency by 28%.
- Implemented DAO patterns and secure database interactions using prepared statements and properties files, enhancing maintainability and scalability.

PROLOGIC FIRST INDIA PVT. LTD., India (*Software Development Intern*) *June 2023 – July 2023*

- Developed a Python chatbot integrating Wit.ai, Snips.ai, and Zapier, improving accuracy by 23% & boosting user engagement.
- Enhanced accessibility with speech-to-text conversion via MediaRecorder API, expanding usability for diverse users.

UW–MADISON CDIS, Madison, WI *(Peer Mentor – AI and DBMS)* *January 2025 – Present*

- Mentored 600+ students in AI & DBMS courses, covering search, ML basics, SQL, schema design, and query optimization.

INFORMATICS SKUNKWORKS, Madison, WI (*Undergraduate Student Researcher)* *Jan 2024 – May 2024*

- Worked under Prof. Dane Morgan to build and evaluate machine learning models using MAST-ML and scikit-learn for engineering datasets, applying cross-validation and hyperparameter tuning, which improved model accuracy and reliability.
- Delivered weekly reports on model improvement, ethical data cleaning, and reproducibility using Python and Google Collab, enhancing team understanding and ensuring data integrity.

PROJECTS

- **Puzzle Lab - Crossword Puzzle Web App (Full-Stack Developer) –** Built using React (Next.js, TypeScript) and Quarkus (Java) with JWT authentication, REST APIs, and MySQL backend. Enabled users to create, edit, and play puzzles with personal accounts.
- **LLM Web Task Evaluation Framework (Machine Learning Engineer) –** Built a Python-based framework to benchmark LLMs on web tasks, evaluating accuracy, robustness, and failure modes via prompt-driven pipelines.
- **NOMinate - Smart Meal Planning App (Full-Stack Developer) –** Developed an intelligent meal recommendation app using OpenAI API, Python, and React, leveraging user preferences and real-time feedback to deliver personalized, non-repetitive suggestions.
- **NYC Taxi Analysis (Data Engineer) –** Developed a Databricks-based ETL and analytics pipeline for NYC Taxi data, leveraging PySpark, Delta Lake, and MLlib for predictive modelling and visualization.
- **RL - Powered Trading Strategy Optimizer (Machine Learning Engineer) –** Created a DQN-based RL agent to optimize trading strategies, boosting simulated returns by 25% and improving accuracy by 30% using Yahoo Finance data.

EXTRACURRICULAR ACTIVITIES

- Active member of DotData Club and Software Development Club at UW–Madison, competed in Kaggle competitions and university hackathons, applying AI/ML and software development skills through coding challenges and technical workshops.